

Review Response Tracker

Objectives & Motivation

#	Reviewers	Feedback	Response in Revision	Checked
1	R1, R3, Brian	Objectives/rationale unclear; novelty and strengths underarticulated	Intro rewritten with 3 new contributions (Sec 1, lines 104–109); Highlights section; Abstract opens with adaptation problem.	✓
2	Brian	State labelled-data scarcity explicitly in problem framing	Sec 1: "require large labelled datasets" (line 98); Sec 2.1: "large labelled tunnel datasets that remain scarce" (line 119)	✓

Literature & Structure

#	Reviewers	Feedback	Response in Revision	Checked
3	AE	Add dedicated Related Works section (not just intro)	Sec 2 with 3 subsections: Sec 2.1 Feature-eng vs DL, Sec 2.2 Foundation-model pipelines, Sec 2.3 LLM reasoning	✓
4	R1, R3, AE, Brian	Structure needs reorganisation; LLM background too long; m/s/k defined too late; pipeline should precede agent details	Restructured: Sec 2 (Related work), Sec 3 (Methodology) with Sec 3.2 (Baseline: SAM4Tun) before Sec 3.3 (R4Tun agents); m/s/k defined in Sec 1 (line 102); Sec 2.3 trimmed	✓
5	AE	Use consecutive numbered reference style starting at [1]	Using natbib with model1-num-names (numbered style)	✓

Experiments & Analysis

#	Reviewers	Feedback	Response in Revision	Checked
6	R2, R3	No ablation analysis to verify component contributions	Cumulative 4-level ablation (Table 2), incremental gains per component (Table 5, Sec 4.2), ablation bar chart (Fig 9)	✓
7	R1, R3	Missing p-values, CIs, effect sizes; small sample size	Paired t-tests, bootstrap 95% CIs, Cohen's d reported throughout; expanded to n=30 tunnels; methodology in Sec 3.4.2; main results in Table 4	✓
8	R2, R3	Limited LLM comparison (was 2 models); no unified variable control	Expanded to 3 LLMs (Opus-4.6, GPT-5.4, Gemini-3-Flash), identical prompts (Sec 3.4.1, Table 3); cross-model summary (Table 6, Sec 4.3)	✓
9	R2	State gives largest gain—need evidence LLM reasoning (not just rules) drives improvement	Sec 4.2: State is the dominant driver.	✓
10	R2	No deterministic rule execution baseline (rules without LLM)	3.4.2. Experimental design: SAM4Tun baseline = deterministic expert rules; run fixed parameters on all 30 tunnels	✓
11	R2	Report complete parameter set with defaults/bounds and LLM config	Baseline parameters for all 4 stages in App A (Tables 8–11); LLM config in Table 3 (model, tokens, temp, timeout)	✓
12	R2	Report per-class IoU, runtime, LLM call count	Per-class IoU in App F (Tables 13–14); practical metrics incl. runtime & API calls in App G	✓

#	Reviewers	Feedback	Response in Revision	Checked
			(Table 15); performance distribution in App E (Table 12)	
13	R2	Sensitivity analysis and parameter robustness	CV analysis of 1,350 parameter files (Sec 3.4.3, Eq 2); 18 critical parameters identified (Table 7, Sec 4.4); tunnel-responsive vs baseline corrections	✓

Figures & Presentation

#	Reviewers	Feedback	Response in Revision	Checked
14	R1, R3	Add comprehensive flow diagram, CoT visualisations, ablation curves	Stage figures (Figs 2–5), architecture (Fig 7), agent design (Fig 8), ablation bar (Fig 9)	✓
15	R2	Figures are blurred and labels unreadable	Regenerated all figures as high-quality vector PDFs	✓
16	R3, AE	Figure placement disrupts flow; use 2-column journal template	Adopted cas-dc 2-column template; figures placed near first reference	✓

Writing & Terminology

#	Reviewers	Feedback	Response in Revision	Checked
17	Brian	Terminology inconsistent ("structured reasoning layer" vs "single-step reflective agent")	Consistent "stage agents" + "context (m, s, k)" throughout text and figures	✓
18	Brian	Figures read as full setup, not examples—mark examples clearly	App C: "condensed excerpt"; App D: "excerpts illustrate"; Sec 3.3 references appendices as sample excerpts	full examples are very long
19	R1, R2, R3, Brian	Language editing needed; reasoning description too abstract; technical wording needs simplifying	Rewriting; CoT anchoring in plain change to "referencing"(Sec 3.3.3);	in progress

Limitations

#	Reviewers	Feedback	Response in Revision	Checked
20	R1, R3	Limitations not specific; no user study for transparency claims	Specific limitations in Sec 5.4: single pipeline/dataset, single reference, no user study, single forward pass; framed as future directions	✓

Supplementary Material

#	Reviewers	Feedback	Response in Revision	Checked
21	R1, R2, R3	Insufficient replicability: missing CoT traces, prompts, implementation logic	CoT worked example (App C, Table), context excerpts (App D, Table), full parameter tables (App A), LLM config (Table 3)	✓
22	AE	Include supplementary material as appendices, not separate	7 appendices (A–G) in appendices.tex, included via \input{appendices}	✓